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(An Autonomous Institute Affiliated to Savitribai Phule Pune University)
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Department of Artificial Intelligence & Data Science

Program name :- Artificial Intelligence & Data Science **Class: S.Y. B. Tech Div:- B**
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Course Name:- Digital Finance
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LoanAI – Financial Decision Support

Project Description :-

LoanAI is a modern fintech decision-support web platform designed to help borrowers make smarter financial decisions. Presented as a next-generation BFSI financial intelligence prototype, it combines four integrated decision engines: loan eligibility assessment, credit score analysis, EMI calculation and AI-based financial guidance, all within a single dashboard-style interface.

The platform evaluates loan eligibility using deterministic underwriting rules such as FOIR and DTI ratios, debt-service coverage and maximum borrowing capacity. It diagnoses credit health through a CIBIL/FICO-style analyzer covering payment history, credit utilization and trade-line age, and calculates reducing-balance EMIs with interactive charts, amortization tables and prepayment savings analysis.

The AI Financial Tips module is prepared for Anthropic Claude integration, using a contextual generative engine to deliver tailored 30/60/90-day advisory roadmaps and financial stress testing. The application uses a clean, dark-themed responsive interface with simple navigation (Home, Loan Eligibility, Credit Score, EMI Calculator, AI Financial Tips), so users can move between tools easily and receive guidance grounded in transparent institutional rules.

Technology & AI :-

Frontend: Responsive dark-themed web interface with interactive SVG donut charts and what-if simulators

Backend / Logic: Rule-based underwriting engine for FOIR/DTI checks, debt-service coverage and borrowing capacity

Credit Analysis: CIBIL/FICO-style diagnostic using payment history, credit utilization and trade-line age

EMI Engine: Reducing-balance interest calculation with annual amortization tables and prepayment savings analysis

AI Integration: Anthropic Claude-ready contextual engine for 30/60/90-day roadmaps and stress testing

Domain: BFSI (Banking, Financial Services & Insurance) decision support

Key Features :-

- Loan Eligibility Checker using deterministic FOIR/DTI assessment
- Debt-service coverage ratio and maximum borrowing capacity classification
- Credit Score Analyzer (CIBIL/FICO) covering payment history, utilization and trade-line age
- Interactive what-if credit score simulations
- EMI Calculator based on reducing-balance interest with real-time interactive SVG donut charts
- Annual amortization tables and prepayment savings analysis
- AI Financial Tips with contextual 30/60/90-day advisory roadmaps and stress testing
- Modern responsive dark-themed interface with easy navigation and quick-action buttons

Scenarios :-

Scenario 1: A borrower opens LoanAI and enters income, obligations and loan details in the Loan Eligibility Checker. The system calculates FOIR/DTI ratios and shows whether the loan is likely to be approved and the maximum amount that can be borrowed.

Scenario 2: A user opens the Credit Score Analyzer to review payment history, credit utilization and trade-line age. Using the what-if simulator, the user sees how changes in behavior could improve the credit score.

Scenario 3: A user plans a loan in the EMI Calculator, entering amount, interest rate and tenure. The tool displays the monthly EMI, a donut chart of principal versus interest, an amortization table and the savings from prepayment.

Scenario 4: Using the AI Financial Tips module, a user receives a personalised 30/60/90-day roadmap and stress-test insights based on their financial profile, helping them plan repayments and improve financial health.

Milestones :-

Milestone 1 – Project Planning & Interface Design

Define the objective of LoanAI, identify the four core decision engines, and design the landing page, navigation and dark-themed dashboard layout.

Milestone 2 – Loan Eligibility Module

Implement deterministic underwriting rules including FOIR/DTI assessment, debt-service coverage and maximum borrowing capacity classification.

Milestone 3 – Credit Score Analyzer & EMI Calculator

Build the CIBIL/FICO-style credit diagnostic with what-if simulations, and the reducing-balance EMI calculator with SVG donut charts, amortization tables and prepayment analysis.

Milestone 4 – AI Financial Guidance

Prepare the Anthropic Claude-ready AI Financial Tips module to generate contextual 30/60/90-day roadmaps and financial stress tests.

Milestone 5 – Testing & Finalisation

Test all four modules with sample inputs, verify calculations, refine the responsive interface and prepare the project for demonstration and documentation.

Conclusion :-

LoanAI brings loan eligibility checking, credit score analysis, EMI calculation and AI-driven financial advice into one platform. By combining transparent, rule-based institutional logic with generative AI guidance, it helps borrowers understand their financial position and make informed borrowing decisions. The project demonstrates how modern web technologies and AI can be applied to BFSI decision support.

Expected Outcome :-

The expected outcome is a working fintech prototype where users can check loan eligibility, analyze credit health, calculate EMIs with visual breakdowns and receive personalised AI-based financial guidance, supporting smarter and more confident financial decisions.